

PYTHON HOMEWORK – ASSIGNMENT 1

WHILE LOOP, BREAK & CONTINUE (EXTENDED)

Learning Objectives

- Master while loops
 - Clearly understand the difference between break and continue
 - Apply logic in real-world style problems
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◆ Part A: Theory & Understanding (Answer in your own words)

1. What is a while loop?
2. When should we use a while loop instead of a for loop?
3. What does break do in a loop?
4. What does continue do in a loop?
5. Can break and continue be used together in the same loop? Explain.
6. What happens if we forget to update the loop condition in a while loop?

◆ Part B: Basic Practice Problems

Problem 1: Counting

Write a program to print numbers from 1 to 20 using a while loop.

Problem 2: Skip Multiples of 3 (Using continue)

Write a program that prints numbers from 1 to 30, but skips numbers divisible by 3 using continue.

Problem 3: Stop at a Number (Using break)

**Write a program that prints numbers starting from 1.
If the number reaches 15, stop the loop using break.**

Problem 4: Sum Until Zero

**Write a program that keeps asking the user for numbers and adds them.
Stop when the user enters 0.
Print the total sum.**

Problem 5: Ignore Invalid Input

Ask the user to enter a number:

- **If the number is negative → skip it using continue**
 - **If the number is greater than 100 → stop the loop using break**
 - **Otherwise → print the number**
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◆ Part C: Break vs Continue (Concept Builder Problems)

Problem 6

Write a program that:

- **Prints numbers from 1 to 10**
- **Skips number 5 using continue**
- **Stops the loop at number 8 using break**

👉 Write the output and explain why it happens.

Problem 7

Explain the difference between these two statements using examples:

- **break**
 - **continue**
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◆ Part D: MINI PROJECTS (EXTENDED)

Mini Project 1: ATM PIN Verification (Using break)

Problem Statement:

- Store a correct ATM PIN in the program
 - Allow the user 3 attempts
 - If the correct PIN is entered:
 - Print “Access Granted”
 - Exit the loop using break
 - If all attempts fail:
 - Print “Card Blocked”
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Mini Project 2: Student Attendance Filter (Using continue)

Problem Statement:

- Ask for student roll numbers
 - Stop when roll number 0 is entered
 - Skip absent students (negative numbers) using continue
 - Print only valid roll numbers
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Mini Project 3: Guess the Number Game

Problem Statement:

- Store a secret number
 - Ask the user to guess the number
 - If the guess is:
 - Wrong → continue asking
 - Correct → print “You Win” and break
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Mini Project 4: Number Classification System

Problem Statement:

- **Continuously take numbers**
- **Skip zero using continue**
- **Stop if number is greater than 100**
- **Print whether the number is:**
 - **Even or Odd**